Algorithms Exercises

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Exercise 1: Sum of List Elements

```
def sum_list(numbers):
    total = 0                  # start with 0
    for num in numbers:        # go through each number
        total += num           # add it to total
    print(total)
```

Exercise 2: Find the Largest Number

```
def find_largest(numbers):
    largest = numbers[0]       # assume first is largest
    for num in numbers:
        if num > largest:      # found a bigger one?
            largest = num      # update largest
    print(largest)
```

Exercise 3: Count Vowels

```
def count_vowels(s):
    vowels = "aeiou"
    count = 0
    for char in s.lower():     # ignore case
        if char in vowels:     # check if it's a vowel
            count += 1
    print(count)
```

Exercise 4: Reverse a String

```
def reverse_string(s):
    reversed_s = ""
    for char in s:
        reversed_s = char + reversed_s        # add each char to the front
    print(reversed_s)
```

Exercise 5: Check Palindrome

```
def is_palindrome(s):
    cleaned = s.lower().replace(" ", "")          # ignore case & spaces
    print(cleaned == cleaned[::-1])               # compare to its reverse
```

Exercise 6: Factorial Calculation

```
def factorial(n):
    result = 1
    for i in range(1, n + 1):                     # multiply 1 x 2 x ... x n
        result *= i
    print(result)
```

Exercise 7: FizzBuzz-style (Dazz/Buzz)

```
for i in range(1, 101):
        if i % 4 == 0 and i % 5 == 0:                 # multiple of both 4 and 5
    print("DazzBuzz")
        elif i % 4 == 0:                              # multiple of 4
    print("Dazz")
        elif i % 5 == 0:                              # multiple of 5
    print("Buzz")
      else:
    print(i)
```

Exercise 8: Find Prime Numbers in a Range

```
def find_primes(start, end):
    primes = []
    for num in range(start, end + 1):
        if num > 1:                                          # primes are > 1
            is_prime = True
            for i in range(2, int(num**0.5) + 1):            # only check up to sqrt(num)
                if num % i == 0:                             # found a divisor
                    is_prime = False
                    break
            if is_prime:
                primes.append(num)
    print(primes)
```

Exercise 9: Fibonacci Sequence

```
def fibonacci(n):
    sequence = [0, 1]
    for i in range(2, n):                                    # build up from the first two
        sequence.append(sequence[-1] + sequence[-2])
    print(sequence[:n])
```

Exercise 10: Calculate the Area of Different Shapes

```
def calculate_area(shape, **dimensions):
    if shape == "circle":
        r = dimensions.get("radius")
        if r is None:
            print("Missing radius")
        else:
            print(3.14159 * r ** 2)
    elif shape == "rectangle":
        l, w = dimensions.get("length"), dimensions.get("width")
        if l is None or w is None:
            print("Missing dimensions")
        else:
            print(l * w)
    elif shape == "triangle":
        b, h = dimensions.get("base"), dimensions.get("height")
        if b is None or h is None:
            print("Missing dimensions")
        else:
            print(0.5 * b * h)
    else:
```

```
    print("Invalid shape")                                    # unknown shape name
```

Exercise 11: Simple Data Aggregation

```
def total_by_language(products):
   totals = {}
   for product in products:                                 # go through each product
          lang = product["language"]
          price = product["price"]
           totals[lang] = totals.get(lang, 0) + price              # add to running total
   print(totals)
```